LUIS RAMIREZ

Mechanical Engineering | New Jersey Institute of Technology | B.S. expected Dec 2027

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Mechanical engineering student with hands-on experience in field engineering and automotive parts, and a growing portfolio of CAD design, FEA/CFD simulation, and mechanism analysis projects from coursework at NJIT. Seven projects are shown below, spanning SolidWorks and Creo Parametric.

Experience: Automotive Parts Specialist, AutoZone (2025 - present) | Field Engineering Technician, Pittenger Paving & Excavating (2021 - 2023) | EOP STEM Tutor, NJIT (2024 - 2025) | Alpha Phi Omega, active brother

COVER LETTER

Luis Ramirez

Dear Hiring Team,

I'm a Mechanical Engineering student at the New Jersey Institute of Technology, graduating in December 2027, and I'm looking for hands-on engineering work where I can learn fast and contribute from day one.

I like turning designs into parts that actually fit and work. In my design course my team modeled an eight-part machine vise in SolidWorks and documented it in a 12-sheet drawing package with sections, details, and a bill of materials, and I checked teammates' sheets. We also validated Creo and SolidWorks simulations of a linkage, a heat sink fin, and a stepped shaft against hand calculations within about 5%. I also designed a self-leveling gyroscopic cup holder, 3D printed a prototype, and revised the fits when it came out too tight, and my portfolio includes a housing cover drawing to ASME Y14.5. At AutoZone, I troubleshoot batteries and electrical parts every shift with testers and OBD-II scanners and explain the fix to customers, and as a Field Engineering Technician at Pittenger Paving & Excavating, I verified layouts on 20+ construction sites while following strict safety procedures. I'm fluent in English and Spanish and comfortable working directly with technicians and production people. I'm eager to learn Lean, Six Sigma, and AutoCAD on the job.

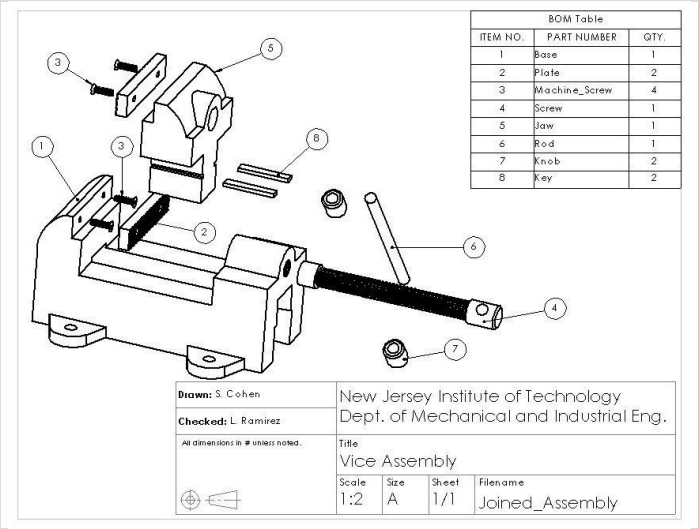
I'd be excited to learn from your team and help with design, testing, and process improvement. Thank you for your time and consideration. I look forward to hearing from you.

Sincerely,
Luis Ramirez

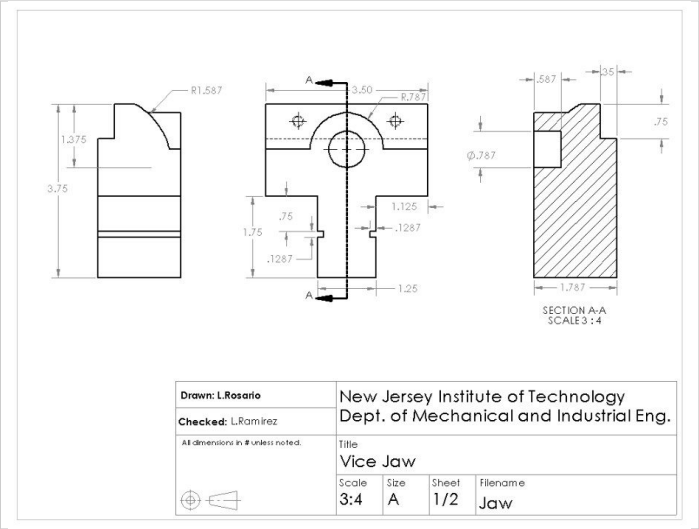
PROJECT 01

Machine Vise CAD & Drawing Package

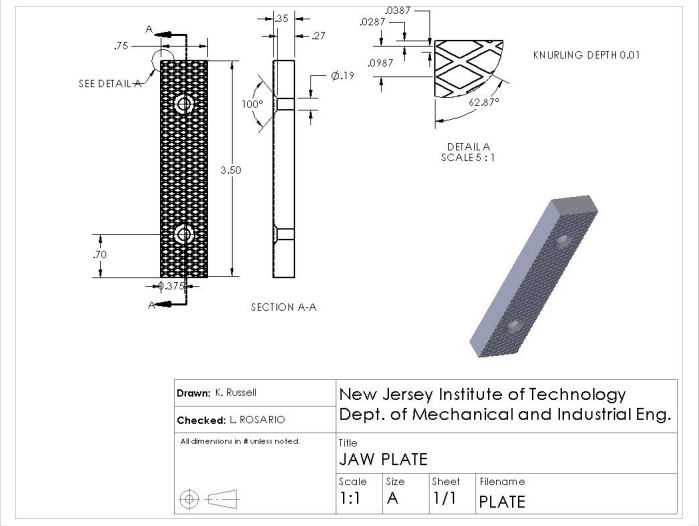
SolidWorks | Team project (ME 403) | ANSI A drawing set



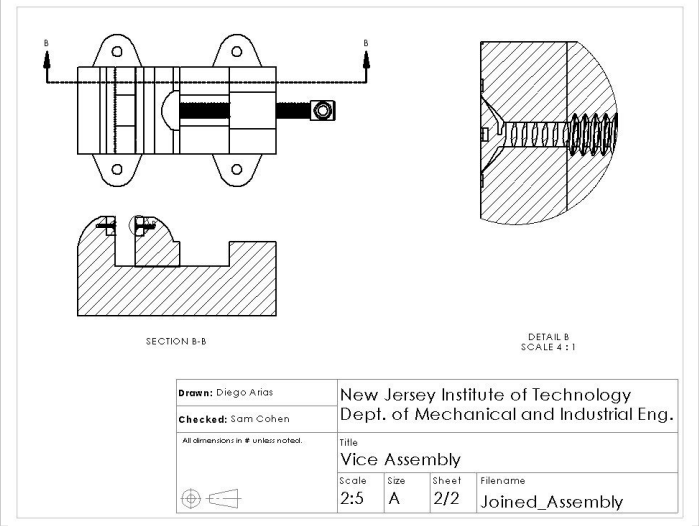
Exploded assembly with balloons and bill of materials.



Movable jaw detail drawing, section A-A (sheet 1 of 2).



Jaw plate with knurl detail (0.01 in depth).



Assembly section through the lead screw and thread detail.

OVERVIEW

An eight-component bench vise modeled in SolidWorks from reference drawings and documented in a 12-sheet ANSI A drawing set so a machinist or inspector could make and check every part from the sheets alone. Each part is a fully constrained parametric model, mated into a working assembly that opens and closes through its full travel, with orthographic, section, and detail views, an exploded view with BOM, and assembly sections.

Skills shown: parametric part and assembly modeling, detail drawings, section and detail views, BOM and balloons, drawing checking, design intent.

KEY RESULTS

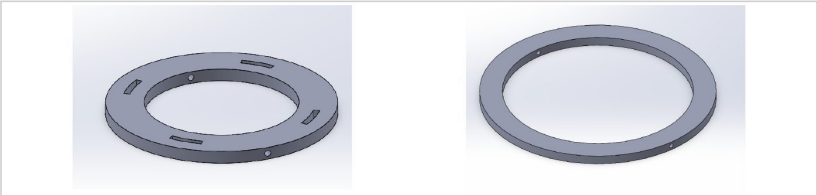
- Base, movable jaw, 2 jaw plates, ACME lead screw, handle rod, 2 knobs, screws, keys
- 12 sheets: 9 part sheets, exploded assembly with BOM, 2 assembly section sheets
- Knurled jaw faces, ACME 5/8"-8 thread, 10-24 UNC screw, 0.005 in x 45 deg chamfers
- Jaw keys, guide slots, and screw bore lined up with no rework at assembly
- ID-driven dimensions regenerated parts and drawings without broken references

PROJECT 02

Gyroscopic Cup Holder

SolidWorks | Team project with Ghazi Mustafa | Concept-to-prototype





Inner and outer gimbal rings, modeled in SolidWorks.





Left: assembled PLA prototype, printed for about $10. Right: prototype holding a standard water bottle during hand-tilt testing.

OVERVIEW

A self-leveling beverage mount for cyclists and wheelchair users, built around a two-ring gimbal: an inner ring cradles the cup and an outer ring lets it pivot about a second axis, so gravity keeps the cup upright as the mounting platform tilts. Three market products were benchmarked before finalizing the concept, and the design was carried from sketch through SolidWorks CAD to a printed and tested prototype.

Skills shown: concept sketching, market benchmarking, multi-part CAD assembly, 3D-print prototyping, design-for-tolerance iteration.

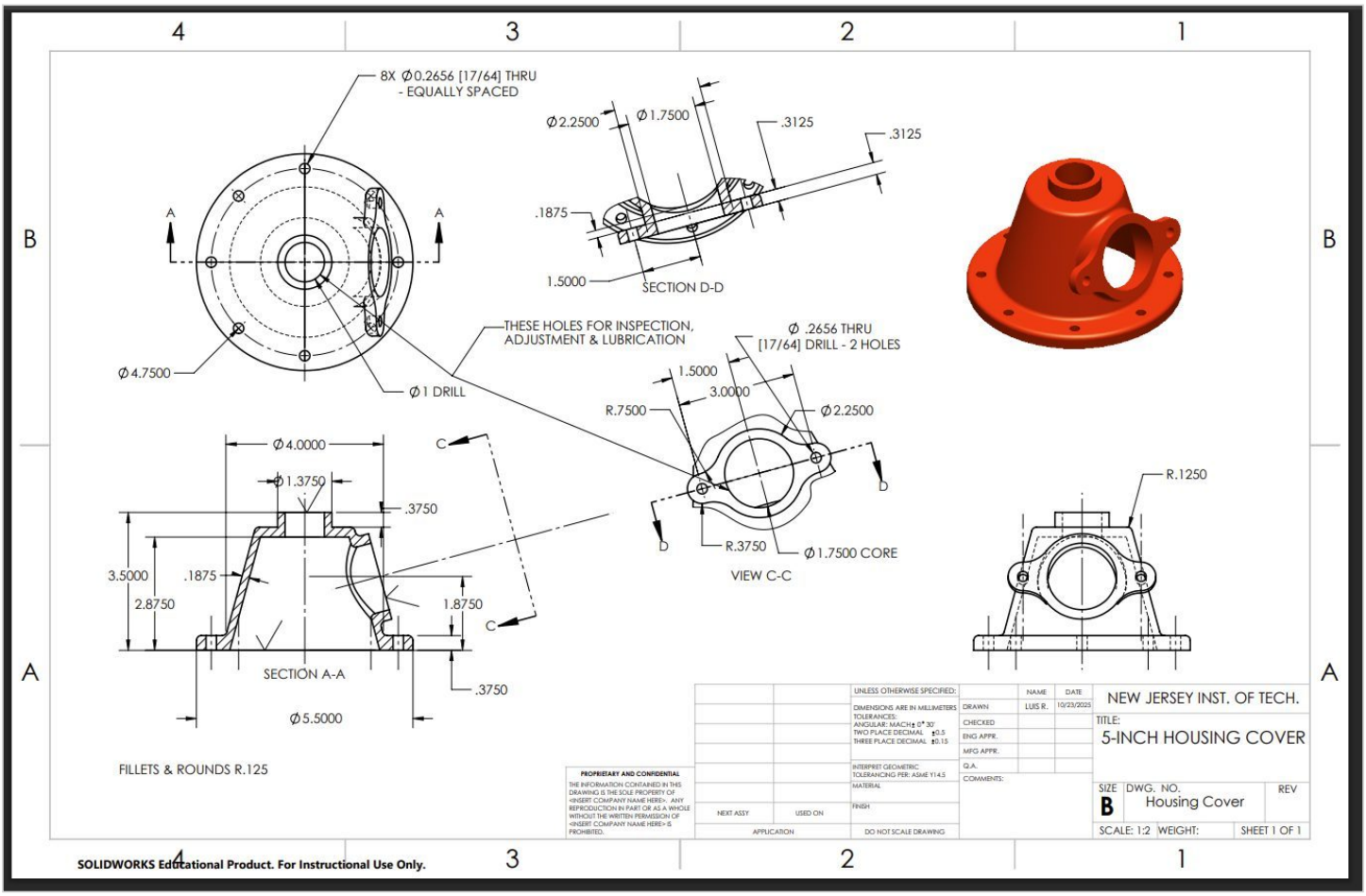
KEY RESULTS

- Fits standard cups 64-100 mm dia.; total weight under 1 kg
- Passive, no electronics; prototype cost approx. $10 in PLA
- Prototype testing found zero-clearance fits too tight
- Revised pin-hole dia. 0.175 in to 0.2 in and eased rib fit by 5 deg

PROJECT 03

5-Inch Housing Cover

SolidWorks | Multiview detail drawing | ASME Y14.5



Complete detailed drawing: top view, Section A-A, Section D-D, auxiliary View C-C, right side view, and shaded isometric.

OVERVIEW

Built a manufacturing-ready detailed drawing of a bell-shaped housing cover from a supplied SolidWorks part model. Six views on a single Size-B sheet at 1:2 scale - a top view, a full section, a projected side view, a full auxiliary section, a partial auxiliary view, and a shaded isometric - fully and unambiguously define the part's stepped bore, mounting-hole pattern, and side-lug geometry per ASME Y14.5 conventions.

Skills shown: multiview and section-view detailing, auxiliary views, GD&T per ASME Y14.5, drawing-package standardization.

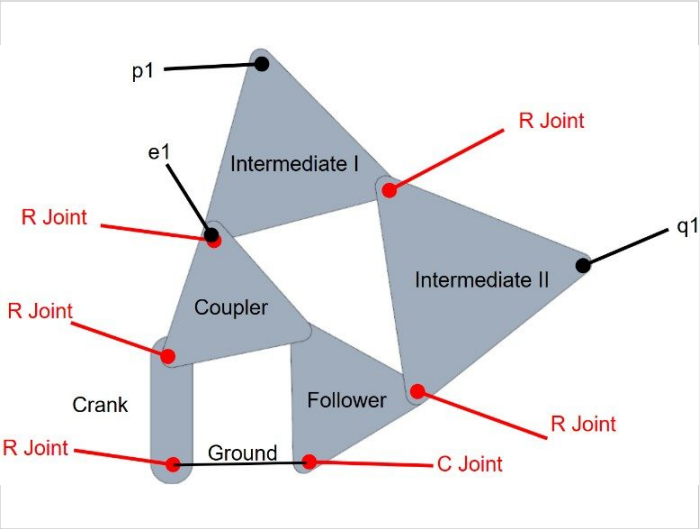
KEY SPECS

- Base flange dia. 5.500 in on a 4.750 in bolt circle
- 8x 17/64 in mounting holes, equally spaced
- Stepped central bore: dia. 2.2500 / 1.7500 in
- General fillets & rounds R0.125 in

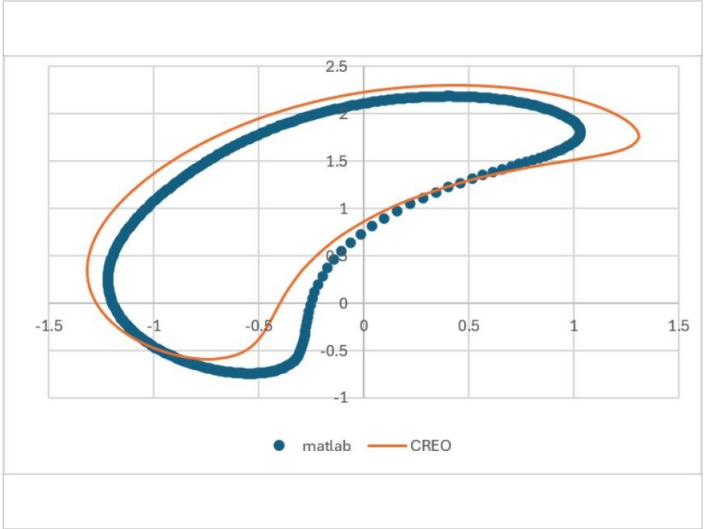
PROJECT 04

Mechanism, Thermal & Structural Validation

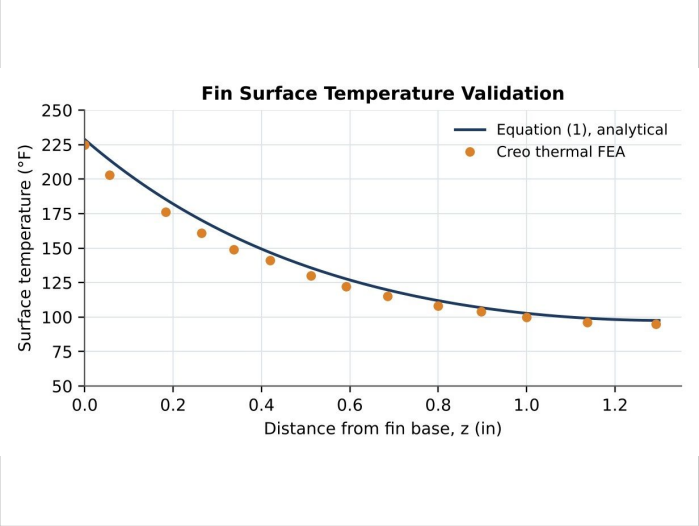
Creo | SolidWorks Simulation | MATLAB | Team project (ME 403)



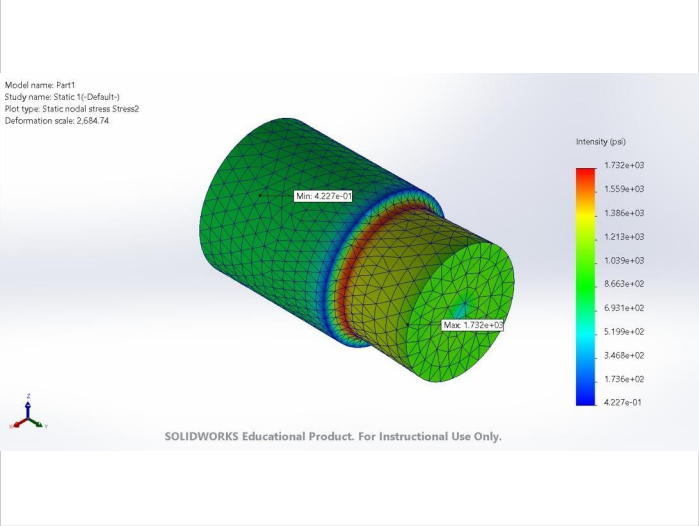
Six-link Watt I mechanism modeled in Creo.



Four-bar coupler curve: MATLAB model vs. Creo.



Fin surface temperature: analytical equation vs. Creo thermal FEA.



Stepped shaft in torsion: SolidWorks stress plot at the fillet.

OVERVIEW

Three CAD-based analyses, each checked against a closed-form model before being trusted: a six-link Watt I linkage path-generation study in Creo, a finned aluminum heat sink with a single-fin conduction-convection validation, and a stepped steel shaft in torsion with a fillet stress concentration solved in SolidWorks Simulation.

Skills shown: mechanism simulation, thermal FEA, stress-concentration FEA, MATLAB analytical models, model validation against hand calculations.

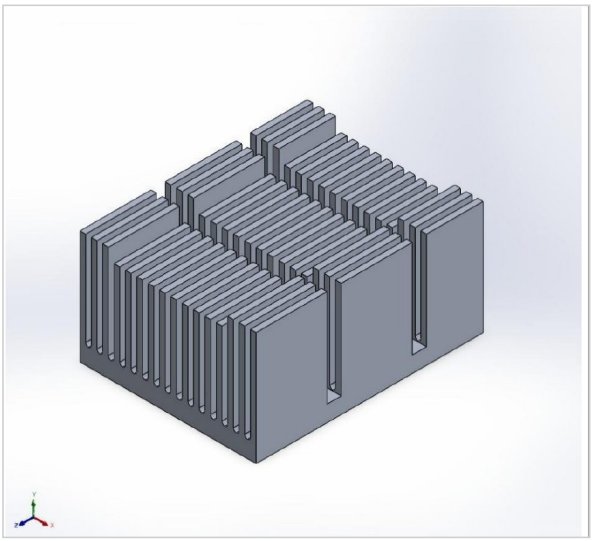
KEY RESULTS

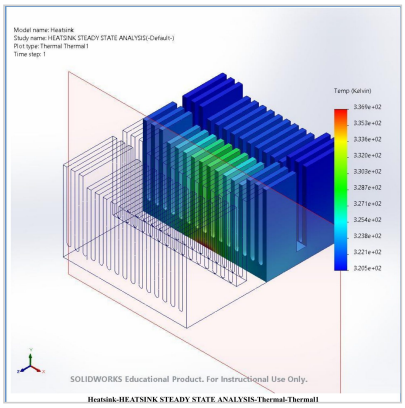
- Creo coupler curve matched the MATLAB displacement model for the four-bar loop
- Fin temperatures fell from 228.7 F at the base to about 97 F at the tip
- FEA within 3-11 F (about 5%) of the analytical fin equation
- Shaft peak shear 866 psi (FEA) vs. 826 psi (Kts hand calculation): 4.8% difference
- Peak stress sat in a ring at the fillet, where the stress-concentration factor predicts it

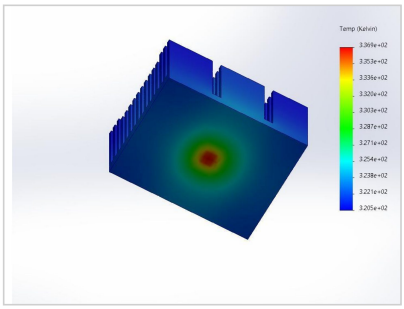
PROJECT 05

CPU Heatsink Thermal Analysis

SolidWorks Simulation | Steady-state thermal FEA | 1060 aluminum







Side section (left) and underside (right) temperature plots. Heat spreads from a central 50 W CPU contact patch, outward through the baseplate and up the fin array.

OVERVIEW

Modeled an extruded finned aluminum heatsink and ran a steady-state thermal FEA with a 50 W CPU heat load on the base and natural convection (h = 25 W/m^2-K) across all 239 exposed faces. The study characterized the temperature gradient from the CPU contact patch to the fin tips and identified which fin channels run warmest, informing where forced-air cooling would help most.

Skills shown: steady-state thermal FEA, convection boundary conditions, heat-flux interpretation, design guidance from simulation results.

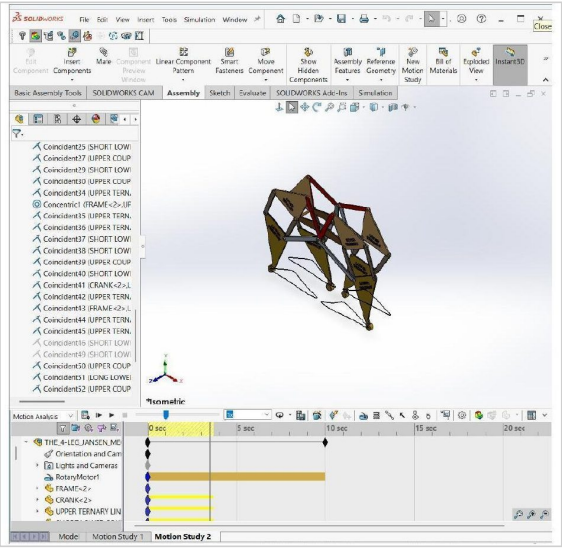
KEY RESULTS

- Max steady-state temp 336.9 K (63.9 C) at the CPU contact patch
- Min temp 320.5 K (47.5 C) at the outer fin tips; delta-T 16.4 K
- Stays well below the 85-100 C silicon-throttling range
- Center fin channels run hottest - best target for a cooling fan

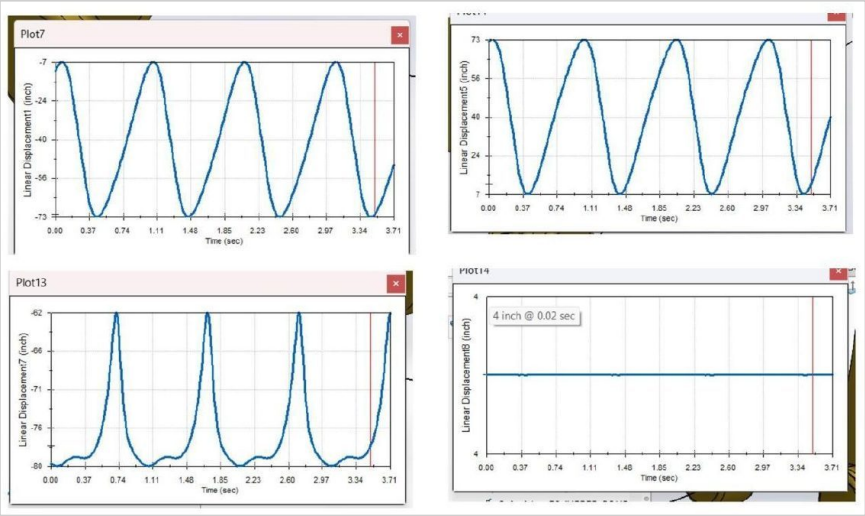
PROJECT 06

4-Leg Jansen Walking Mechanism

SolidWorks Motion | Planar eight-bar linkage | Kinematic simulation



Four-leg Jansen assembly mid-simulation in SolidWorks, with the mate list and RotaryMotor1 Motion Study timeline visible.



Linear displacement time histories: X-displacement for two legs shows an out-of-phase walking gait; Y-displacement (bottom-left) shows the flat stance / fast swing foot curve; Z stays flat, confirming planar motion.

OVERVIEW

Assembled a four-legged walking base built on the Jansen linkage, a planar eight-bar leg mechanism whose single rotating crank produces a foot-shaped path with no separate ankle or gait controller. Drove the assembly with a rotary motor in SolidWorks Motion at 60 RPM and extracted quantitative displacement/velocity time histories to back up the animation.

Skills shown: mechanism/linkage assembly, motor-driven motion studies, trace-curve and time-history data extraction, kinematic validation.

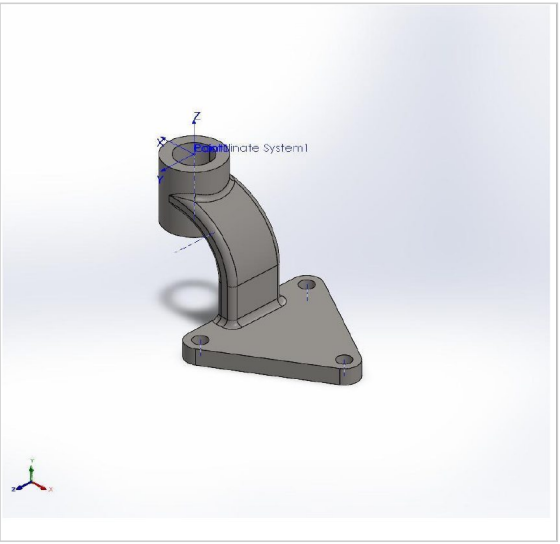
KEY RESULTS

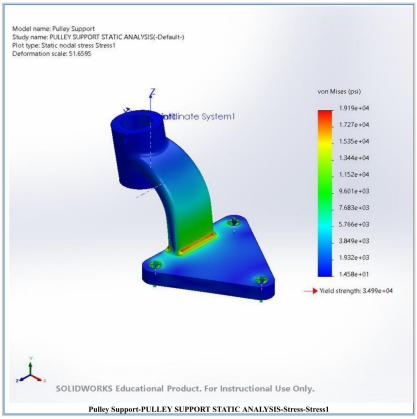
- ~20 leg-link instances plus shared frame/crank, all 1-DOF driven
- Y-displacement shows the signature flat-stance/fast-swing gait
- Two legs run exactly out of phase, confirming a stable 4-leg gait
- Z-displacement/velocity ~0, confirming fully planar motion

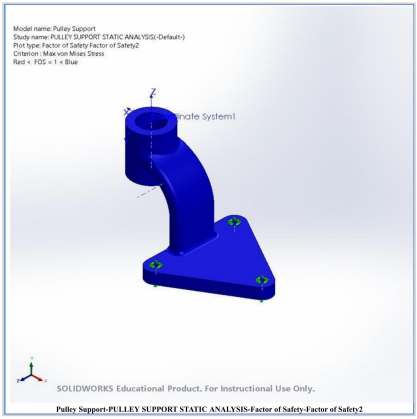
PROJECT 07

Mechanical Drive Pulley Support

SolidWorks Simulation | Static FEA | Cast alloy steel bracket







Von Mises stress (left) peaks at the base fillet; Factor-of-Safety plot (right) shows the entire bracket above FOS = 1.0.

OVERVIEW

Ran a static structural FEA on a cast-alloy-steel pulley support bracket under a 100 lbf horizontal bearing load and a 20 lbf vertical load, with the three base bolt holes fixed. A refined mesh (0.033 in elements) was applied at the base fillets to resolve stress concentrations, and a Design Insight study identified low-stress material that could be removed in a future light-weighted revision.

Skills shown: static FEA setup, mesh refinement, stress/FOS/displacement interpretation, topology-informed light-weighting.

KEY RESULTS

- Max von Mises stress 19,190 psi vs. 34,994 psi yield (FOS 1.82)
- Max deflection 0.0093 in at the top of the bearing collar
- Mesh: 50,260 nodes / 32,645 elements, 0% distorted
- Design Insight: over 65% of material carries little to no load